October 14, 2022

APS110 – Lecture 15

See hand written notes for drawings.

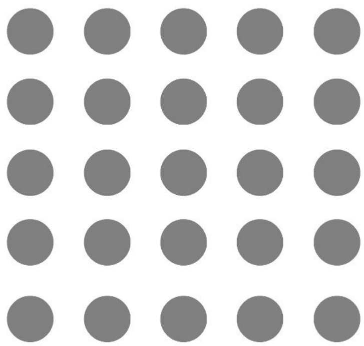
Imperfections

- "defects" is another term used for imperfections → can imply that there's something *wrong*, which is not true as imperfections are very important for changing the qualities of materials
- Through imperfection populations, we as engineers can control the properties of materials
- We're going to begin by looking at **strengthening metals** (changing mechanical properties) through these imperfections
- An imperfection is a disruption in the regular arrangement of atoms

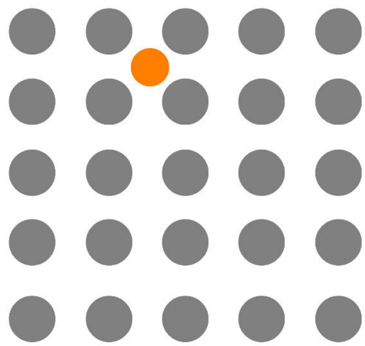
How can we categorize these imperfections?
→ Number of dimensions that define the imperfection

- <u>0 Dimensional</u> → **Point Imperfections**
  - <u>Ex</u>. impurities (dissolved)
- <u>1 Dimensional</u> → **Linear Imperfections**
  - <u>Ex</u>. dislocation
- <u>2 Dimensional</u> → **Interfacial Imperfections**
  - <u>Ex</u>. grain boundary, free surface
- <u>3 Dimensional</u> → **Volume Imperfections**
  - <u>Ex</u>. second phase

Zero Dimensional (Point Imperfections)



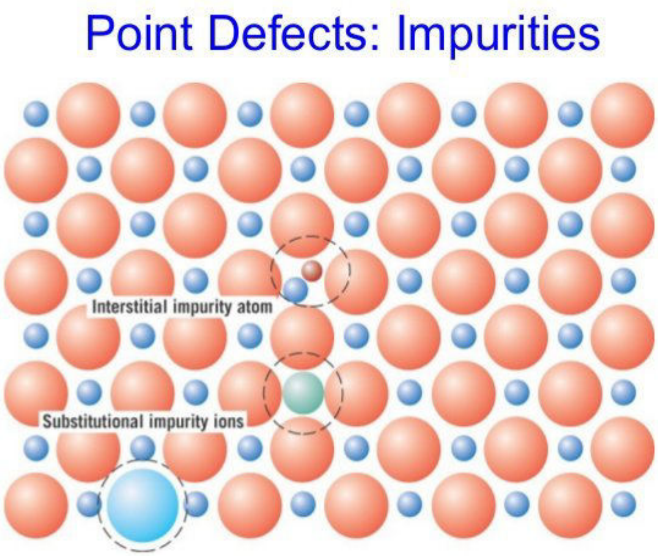
Perfect Crystal

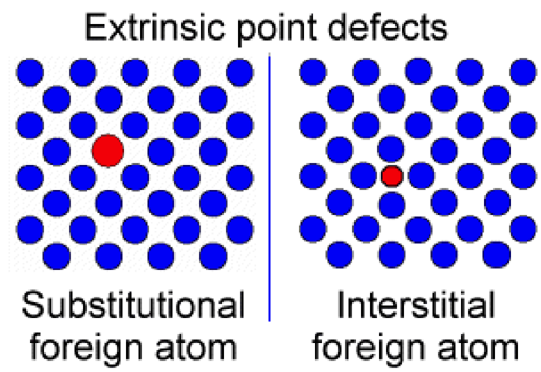


Interstitial Atom - Point Defect

- Ex. Carbon (C) in Iron (Fe) → interstitial impurity
  - Carbon is small and fits nicely between the iron atoms
  - For an atom to dissolve into an interstitial site, it needs to be quite a bit smaller than the lattice atoms (50% or less)
- What happens when we try to dissolve a larger atom?
  - Substitutional impurity → takes the place of another atom in the lattice

        - Ex. Copper (Cu) in Nickel (Ni)





- With point imperfections, we're <u>not</u> changing the crystal structure, we're just substitution one atom for another
- There is a small threshold for how much impurity a lattice can accommodate (because of lattice strain, electronegativity, etc.) before there will be a change of structure (compound formation)
    - Ex. iron can only accommodate 0.02 wt % carbon dissolved

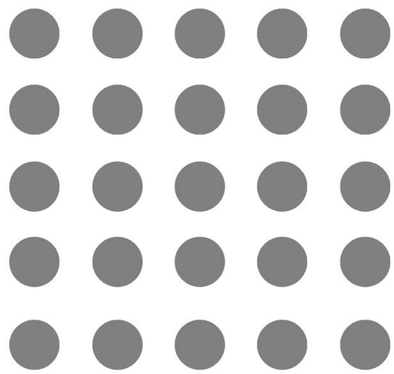
Alloy – a metal with impurities added (a metal with some amount of a second metal added)
- Could include point imperfections (dissolved impurities) <u>OR</u> volume imperfections
    - Could be both
    - These types of imperfections interact slightly differently

Compound – a substance with more than one element and a specific stoichiometry
- Ex. Fe_3C, Al_2O_3
- Have a specific chemical formula

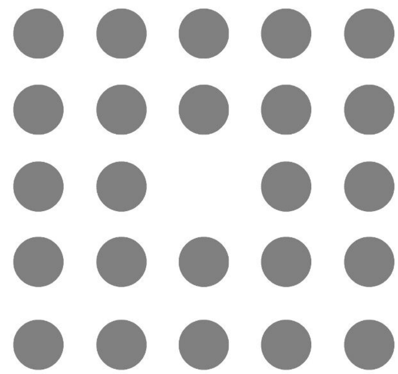
<u>Recall</u>: dissolved impurities can inhibit dislocation movement → strengthening

Another type of point imperfection: **Vacancy**
- Missing atom
- Not that useful for us to change the mechanical properties

October 14, 2022



Perfect Crystal



Vacancy - Point Defect

- Unsatisfied bonds with the neighbour atoms
- Vacancies exist in ceramics as well
- These imperfections are ALWAYS present, and thermally generated
  - Atoms are always vibrating around, and they're trying to make a jump away from its nearest neighbour atoms
  - Occasionally one atom has enough kinetic energy to move somewhere else, and it leaves behind this empty spot
- Battle between the binding energy (holding the atom in place, bond energy) and the thermal energy
- If the temperature is higher, we'll get more vacancies, as the atoms in the material have more energy and more atoms will be able to break free of their binding
- Number of vacancies, $\mathbf{N_v = N e^{(-Qv/kT)}}$
  - N – total lattice sites
  - Q_v – energy for vacancy formation
  - kT – thermal energy (Boltzmann Constant * absolute temperature)